

Bounding the external field in mean-field opinion dynamics: what any content observable must satisfy to drive a transition

Brian Mwai

Abstract

Mean-field opinion models borrowed from the Ising universality class are routinely used to argue that media content can tip a population between competing consensus states. In almost every case the external field representing that content is assigned by hand. The models therefore demonstrate that a field of sufficient magnitude tips the population, while saying nothing about whether any real content observable reaches that magnitude. We close that gap from the theory side. Writing the field as $h = \alpha X$ for an arbitrary content observable X with attainable range ΔX , we show that no media-driven transition is possible unless the coupling satisfies $\alpha \geq h_c(\beta J)/\Delta X$, where $h_c(\beta J)$ is the spinodal field at which the metastable minimum disappears. The bound inverts the usual procedure: instead of fitting α and hoping the estimate clears zero, it states what α would have to be for the mechanism to operate at all, and it can be evaluated before any data are collected. We derive the Landau coefficients from the self-consistency condition rather than asserting them, recover the mean-field exponents $\beta = 1/2$, $\gamma = 1$ and $\delta = 3$ numerically as a check on the solver, and give a screening criterion that any proposed field observable must pass.

Keywords: sociophysics, opinion dynamics, mean-field Ising, Landau theory, phase transition

1. Introduction

The statistical physics of social dynamics has a standard move: identify a binary attitude with an Ising spin, identify social conformity with a ferromagnetic coupling J , identify media or propaganda with an external field h ,

and study when the population's average opinion m jumps between consensus states [1, 2]. The move is productive. It supplies an order parameter, a control parameter and a transition, and it explains why small changes in influence can produce disproportionate changes in collective opinion near a critical point.

It also has a weakness that is rarely stated. The field h enters as a free parameter. A study fixes h at whatever value makes the phenomenon visible, sweeps it, and reports that the population tips at some h^* . The pattern is consistent across the literature that models media explicitly: Crokidakis [3] introduces mass media into the two-dimensional Sznajd model as a probability p of following the media opinion and sweeps it, finding the phase transition absent above $p \approx 0.18$; Azhari and Muslim [4] apply an external field to majority-rule and q -voter dynamics and sweep its magnitude; the review by Sirbu et al. [5] surveys models combining peer interaction with external information without any of them measuring the external term. In each case the swept parameter is the quantity whose real-world magnitude is at issue.

Nothing in such an exercise establishes that real media content produces a field anywhere near h^* , because nothing in it connects h to a measurable property of content at all. The conclusion "a sufficiently strong field tips the population" is a statement about the model, not about media.

This paper takes the missing step in the direction that does not require data. Suppose an observable X is proposed as the content-side quantity driving the field, through a linear map $h = \alpha X$ with an unknown coupling α . Suppose further, as is true of any real observable, that X has a bounded attainable range ΔX across the content a population actually encounters. Then the requirement that the field be able to reach the spinodal imposes a lower bound on α that involves no fitting:

$$\alpha \geq \frac{h_c(\beta J)}{\Delta X}. \quad (1)$$

Everything on the right is either a property of the model (h_c as a function of the coupling βJ) or a property of the observable that can be measured cheaply (ΔX). The bound is therefore a screening test: it can reject a proposed mechanism before any expensive measurement of α is attempted, and it converts a failure to detect a coupling into a quantitative statement about how large the coupling would have had to be.

Section 2 fixes the model. Section 3 derives the Landau coefficients from the self-consistency condition, correcting a form that does not follow from it. Section 4 validates the solver against the three exact mean-field exponents. Section 5 derives the bound and Section 6 states the screening criterion.

2. Mean-field model

Take N agents carrying binary opinions $s_i = \pm 1$, coupled all-to-all with strength J/N , in an external field h :

$$\mathcal{H} = -\frac{J}{N} \sum_{i < j} s_i s_j - h \sum_i s_i. \quad (2)$$

The magnetisation $m = N^{-1} \sum_i s_i$ is the collective opinion polarisation, and in the thermodynamic limit it obeys the self-consistency condition

$$m = \tanh(\beta J m + h), \quad (3)$$

with $\beta = 1/k_B T$ and $k_B = 1$ throughout. We follow the convention that h absorbs the factor β , so h is dimensionless and directly comparable to βJ . The social reading of β is the inverse of a noise level: large β describes a population that copies its neighbours reliably, small β one that decides close to at random.

Equation (3) has a single root for $\beta J < 1$ at zero field and three for $\beta J > 1$, of which the outer two are stable. The transition at $\beta J = 1$ is continuous.

3. Landau expansion with derived coefficients

Write the free energy as a quartic in the order parameter,

$$F(m) = a m^2 + b m^4 - h m. \quad (4)$$

The coefficients are not free. Requiring that $\partial F / \partial m = 0$ reproduce (3) to cubic order fixes them. Inverting (3) gives $\text{artanh}(m) = \beta J m + h$, and expanding $\text{artanh}(m) = m + m^3/3 + O(m^5)$ yields

$$(1 - \beta J) m + \frac{1}{3} m^3 - h = 0. \quad (5)$$

Matching term by term against $\partial F/\partial m = 2am + 4bm^3 - h$ gives

$$a = \frac{1 - \beta J}{2}, \quad b = \frac{1}{12}. \quad (6)$$

Both follow from (3); neither is chosen. This is worth stating explicitly because coefficient forms that do *not* follow from the accompanying self-consistency condition appear in the applied literature, and the symptom is subtle: the free-energy curve still has the right qualitative double well, so the figure looks correct, while its minimum sits away from the m^* that the same paper's root-finder reports. The double well opens when $a < 0$, that is when $\beta J > 1$, recovering the transition already implied by (3).

Figure 1 shows (4) at zero field on both sides of the critical coupling.

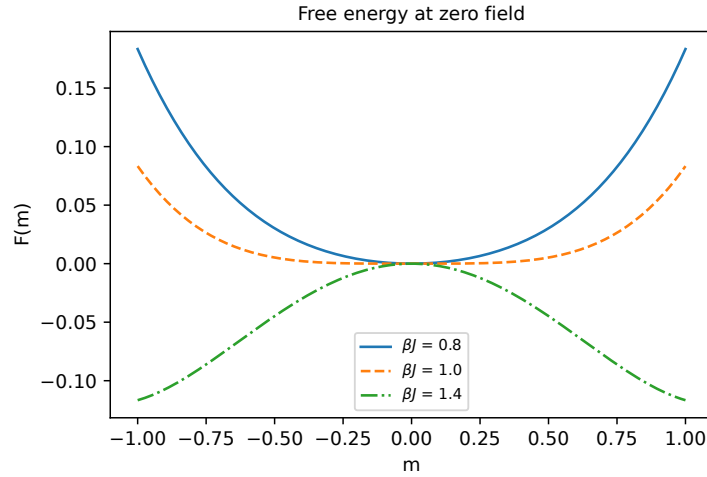


Figure 1: Free energy (4) at zero field for $\beta J = 0.8$, 1.0 and 1.4 with the coefficients (6). The second minimum appears only above $\beta J = 1$.

The linear response to the field is

$$\chi = \frac{\beta \operatorname{sech}^2(\beta J m^* + h)}{1 - \beta J \operatorname{sech}^2(\beta J m^* + h)}, \quad (7)$$

which diverges where the denominator vanishes and is undefined beyond it; we report χ only where the denominator is positive.

4. Numerical validation against the exact exponents

Every later figure is drawn with the same solver, so the solver is checked first against results that are known exactly. Mean-field theory fixes

$$\beta = \frac{1}{2}, \quad \gamma = 1, \quad \delta = 3, \quad (8)$$

for $m^* \sim (\beta J - 1)^\beta$ at zero field, $\chi \sim |1 - \beta J|^{-\gamma}$, and $m \sim h^{1/\delta}$ on the critical isotherm.

One methodological point matters here. Solving (3) by direct iteration of $m_{k+1} = \tanh(\beta J m_k + h)$ is the natural implementation and is adequate away from criticality, but its convergence degrades precisely at $\beta J = 1$, where the map's derivative reaches unity. The exponents are defined in that limit, so fitting them to iterated values measures truncation error rather than physics: doing so returns $\delta \approx 1.35$ against an exact 3. We therefore fit the exponents to roots obtained by bracketing, and check the iterative solver against the bracketed one away from criticality, where the largest absolute discrepancy over the sampled grid is 1.15×10^{-9} .

Fitting inside a window of width 0.05 in $|\beta J - 1|$ at 60 points recovers

$$\beta = 0.4872, \quad \gamma = 1.0000, \quad \delta = 3.0042, \quad (9)$$

with absolute errors 0.0128, 0.0000 and 0.0042 respectively. The residual error in β is the largest of the three and is a finite-window effect: the power law is asymptotic, and the quartic term contaminates the slope as the window widens. Figures 2, 3 and 4 show the three fits.

5. The spinodal field and the bound

Above $\beta J = 1$ the free energy has two minima at zero field. Raising the field deepens one and shallows the other; at a finite field the shallower minimum disappears entirely. That value is the spinodal field $h_c(\beta J)$, and it is the field a population in the metastable state must experience before it is obliged to move. We locate it by bisection on the presence of a second stationary point, which keeps it consistent with the solver used elsewhere rather than introducing a closed form derived under different approximations. Figure 5 shows the resulting boundary; h_c vanishes as $\beta J \rightarrow 1^+$ and grows monotonically, reaching 0.53284 at $\beta J = 2.00$.

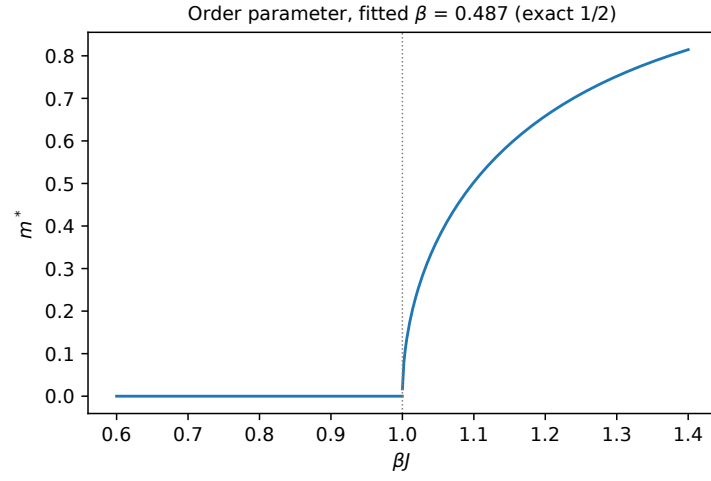


Figure 2: Order parameter across the transition. The fitted exponent is 0.4872 against an exact $1/2$.

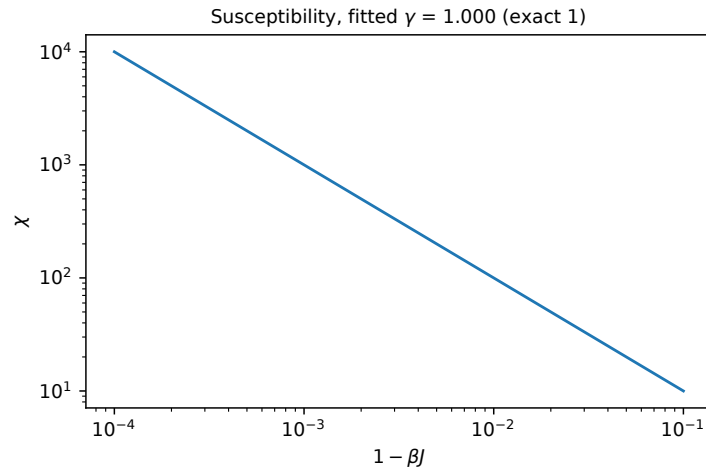


Figure 3: Susceptibility (7) approaching criticality from below, fitted exponent 1.0000 against an exact 1.

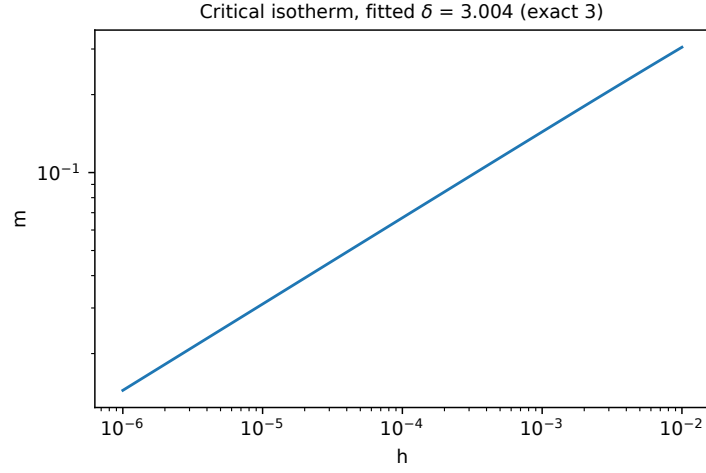


Figure 4: Critical isotherm at $\beta J = 1$, fitted exponent 3.0042 against an exact 3.

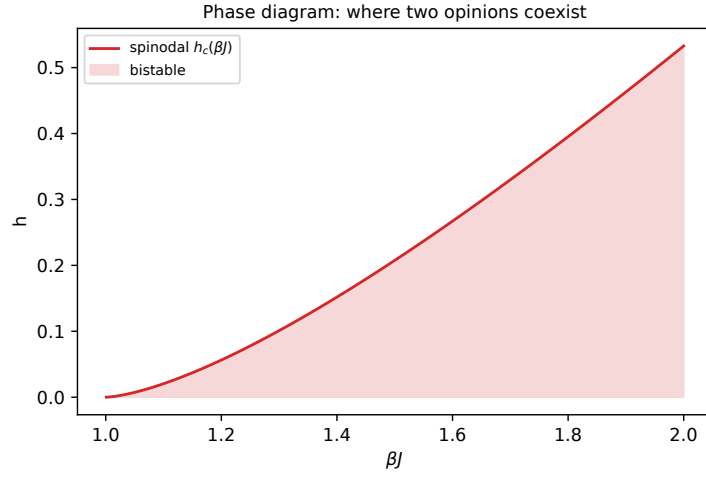


Figure 5: Phase diagram in $(\beta J, h)$. The shaded region is where two opinion states coexist; its upper boundary is the spinodal $h_c(\beta J)$.

Now let the field be generated by a content observable. Any such proposal has the form

$$h = \alpha X, \quad (10)$$

where X is measurable on a piece of content and α is an unknown coupling carrying the units. Real observables are bounded: across the content a population plausibly encounters, X spans some finite range ΔX , which is an empirical property of the observable and of the corpus, not of the model. The largest field the mechanism can deliver is therefore $\alpha\Delta X$. Demanding that this reach the spinodal gives

$$\alpha\Delta X \geq h_c(\beta J) \quad \Longleftrightarrow \quad \alpha \geq \frac{h_c(\beta J)}{\Delta X}. \quad (11)$$

Figure 6 plots the right-hand side against βJ for several values of ΔX . At $\beta J = 2.00$ the required coupling is 5.3284 for an observable of spread 0.1, 1.0657 for spread 0.5, 0.5328 for unit spread and 0.2664 for spread 2.

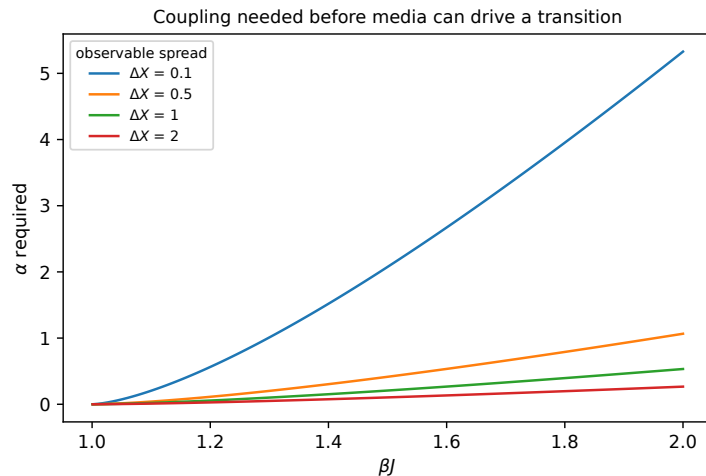


Figure 6: Coupling required before a content observable of spread ΔX can drive the population across the spinodal.

Three properties of (11) are worth drawing out.

First, it is a necessary condition, not a sufficient one. Clearing the bound does not establish that content tips populations; it only fails to rule it out. Falling below it, however, is decisive within the model's assumptions.

Second, it does not require the field to be estimated. ΔX is obtained by measuring the observable across a corpus, which is ordinarily far cheaper

than estimating a coupling to collective behaviour, and h_c comes from the model.

Third, it turns a null result into a quantitative constraint. A study that measures α and obtains an interval containing zero has, on its own, only the statement that no coupling was detected. Combined with (11), the same interval says that the coupling would have to exceed $h_c(\beta J)/\Delta X$ for the mechanism to work, and that the measurement excludes that range at the stated confidence. This is a stronger and more useful claim, and it is available to any study that reports a spread alongside its null.

6. A screening criterion

The bound supports a procedure that can be run before data collection.

1. State the observable X and the map $h = \alpha X$ explicitly, including units.
2. Measure or bound ΔX on a modest sample of the content of interest. This does not require the outcome variable and is usually cheap.
3. Choose the range of βJ the population plausibly occupies and read $h_c(\beta J)$ from Figure 5.
4. Evaluate $\alpha_{\min} = h_c(\beta J)/\Delta X$ and ask whether a coupling of that size is defensible on independent grounds.
5. If it is not, the mechanism is excluded within the model and the expensive measurement is unnecessary. If it is, proceed, and report ΔX alongside the estimate of α so that the bound can be applied to the result.

The weakest link is step 4, since α has no independent scale in most applications. We note that the criterion is still informative in relative terms: two candidate observables can be ranked by $\alpha_{\min}$, and an observable whose spread is small enough to demand an implausibly large coupling is a poor choice of field regardless of how well it correlates with anything.

7. Limitations

The result inherits every assumption of the mean-field model: all-to-all coupling, a single homogeneous population, a static field, and a binary opinion. Network structure, heterogeneous susceptibility and temporal dynamics all change h_c , and in general lower it, so (11) should be read as a bound

within this model rather than a universal one. The linear map $h = \alpha X$ is the simplest possible coupling and is an assumption in its own right; a saturating map would change the relationship between ΔX and the maximum attainable field. Finally, the spinodal is the relevant threshold for a population trapped in a metastable state; a population at the coexistence field can be moved by a smaller field given enough time, so (11) is conservative in the sense of demanding more of α than a nucleation argument would.

8. Reproducibility

All figures and every number quoted in this paper are produced by a single script, `scripts/phase_boundary.py`, which writes both the figures and a JSON record of the values; the numbers in the text are generated from that record rather than transcribed. The exponent check is enforced: the script exits non-zero if any fitted exponent departs from its exact value by more than a stated tolerance, so a solver regression fails the run instead of producing plausible curves.

Data availability

No data were used. The paper is analytic and numerical throughout.

References

- [1] C. Castellano, S. Fortunato, V. Loreto, Statistical physics of social dynamics, *Rev. Mod. Phys.* 81 (2009) 591–646.
- [2] S. Galam, Sociophysics: A review of Galam models, *Int. J. Mod. Phys. C* 19 (2008) 409–440.
- [3] N. Crokidakis, Effects of mass media on opinion spreading in the Sznajd sociophysics model, *Physica A* 391 (2012) 1729. arXiv:1111.5750.
- [4] Azhari, R. Muslim, The external field effect on the opinion formation based on the majority rule and the q -voter models on the complete graph, *Int. J. Mod. Phys. C* (2023) 2350088. arXiv:2305.06051.
- [5] A. Sîrbu, V. Loreto, V. D. P. Servedio, F. Tria, Opinion dynamics: models, extensions and external effects, in: *Participatory Sensing, Opinions and Collective Awareness, Understanding Complex Systems*, Springer, 2016, pp. 363–401. arXiv:1605.06326.

- [6] K. Sznajd-Weron, J. Sznajd, Opinion evolution in closed community, Int. J. Mod. Phys. C 11 (2000) 1157–1165.
- [7] D. Stauffer, Social applications of two-dimensional Ising models, Am. J. Phys. 76 (2008) 470–473.
- [8] N. Goldenfeld, Lectures on Phase Transitions and the Renormalization Group, Addison-Wesley, 1992.